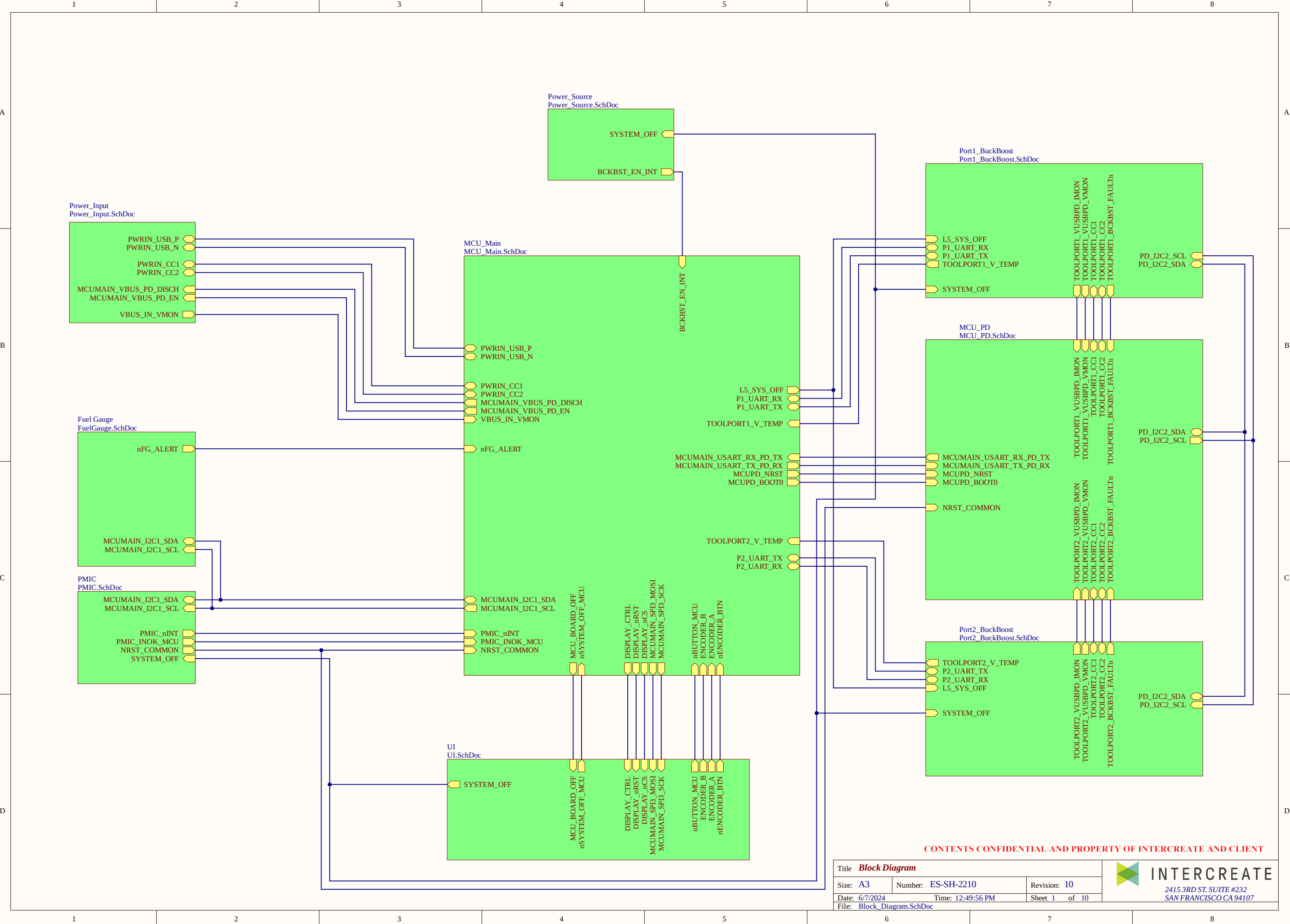
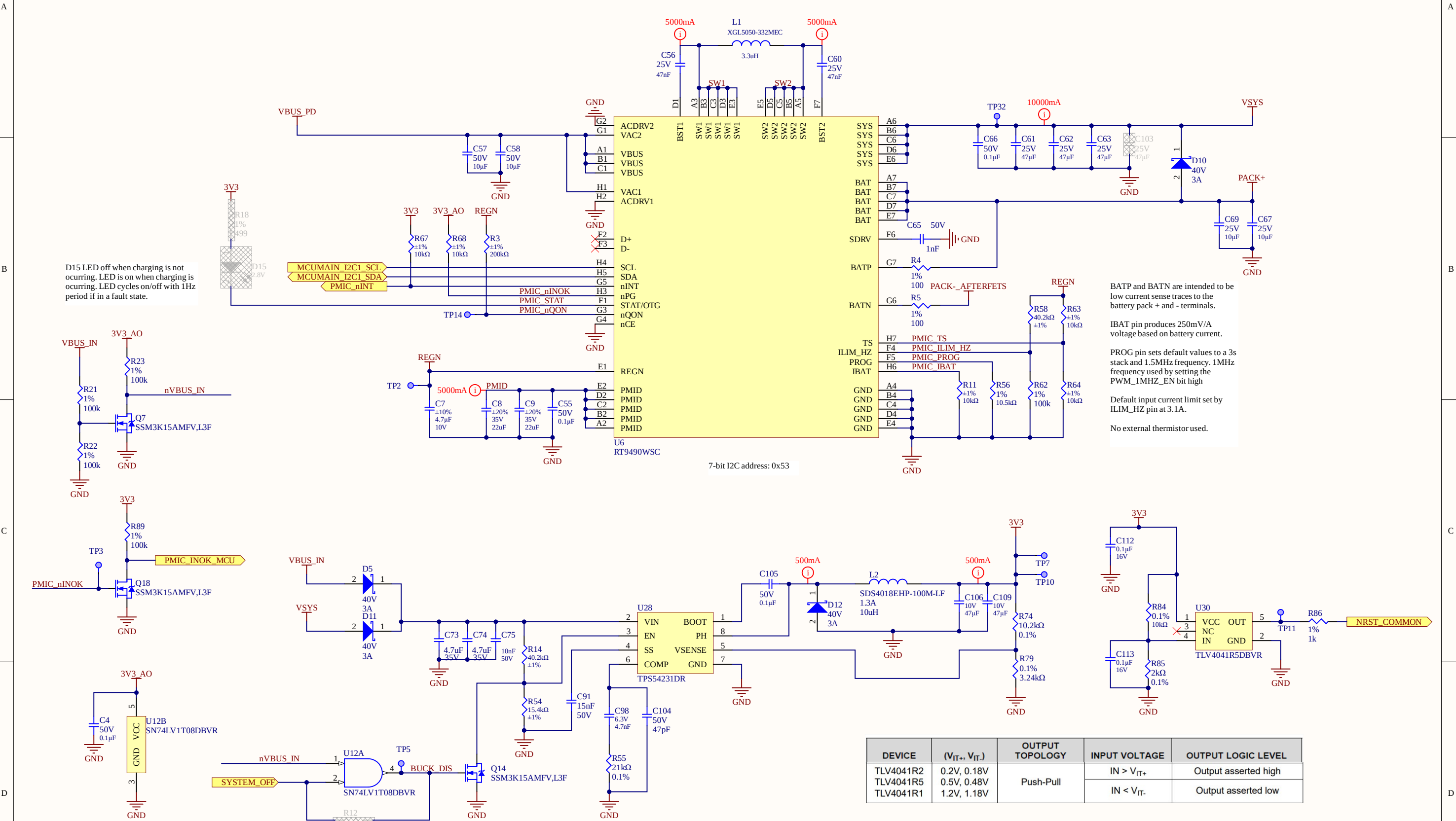
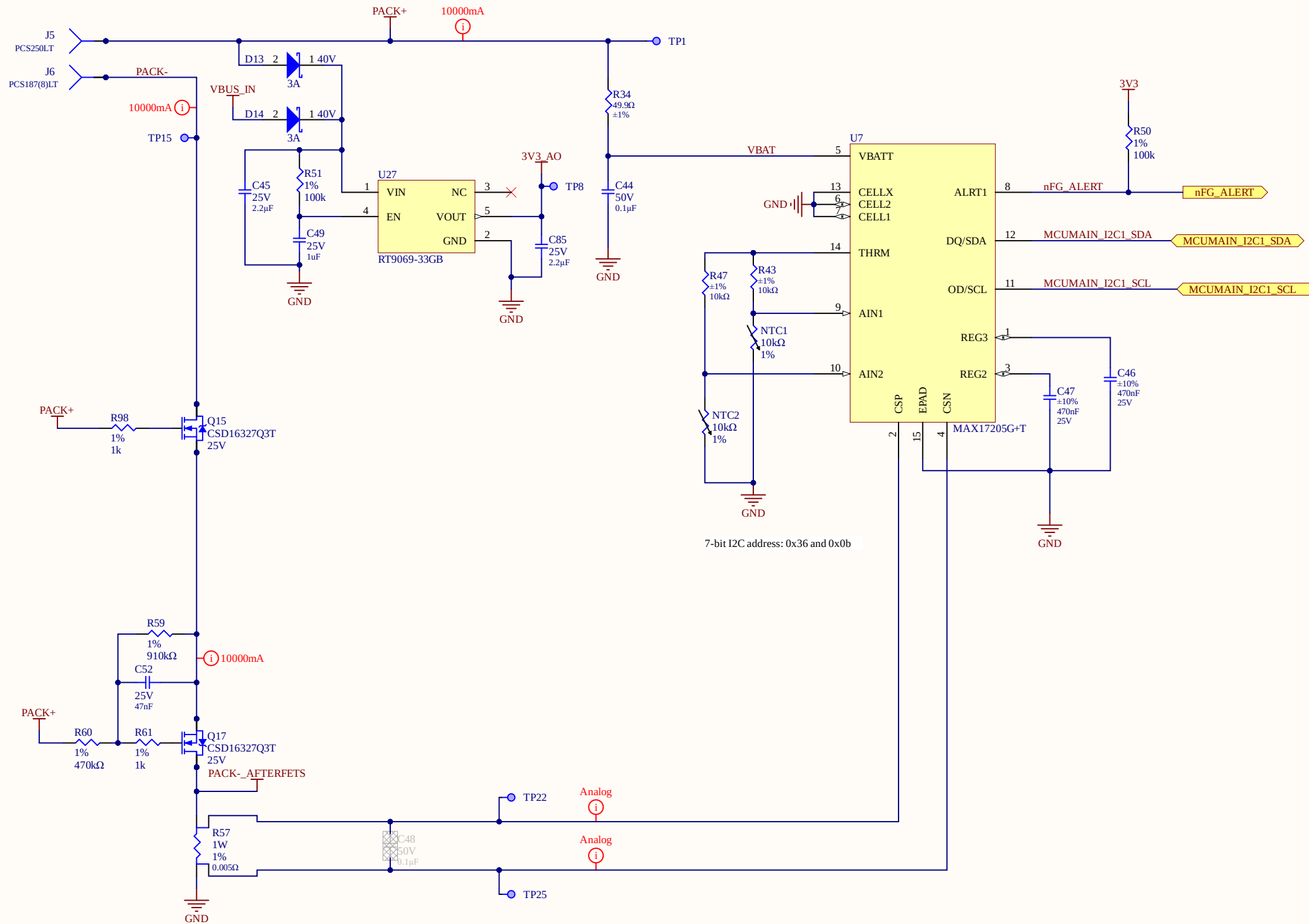






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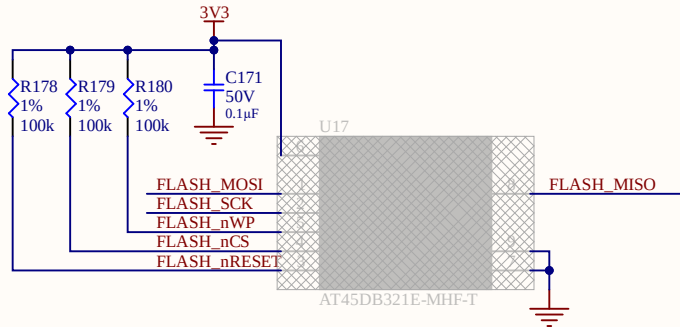
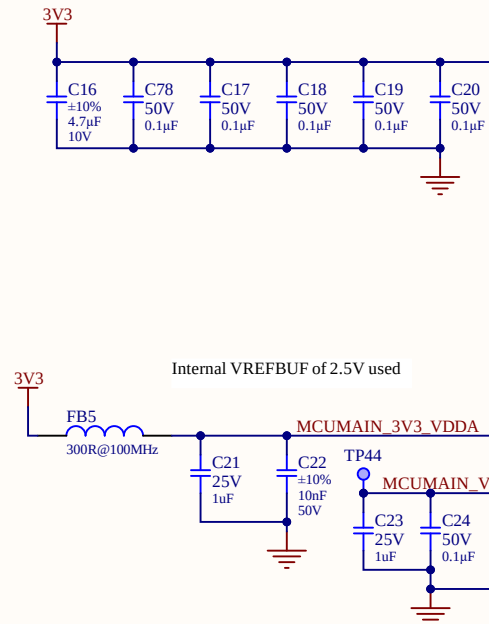
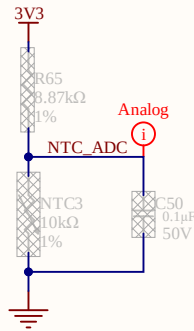
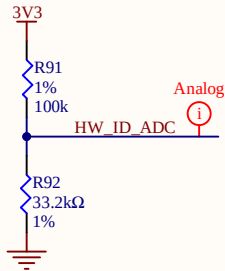


CONTENTS CONFIDENTIAL AND PROPERTY OF INTERCREATE AND CLIENT

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Size: A3	Number: ES-SH-2210	Revision: 10
Date: 6/7/2024	Time: 12:49:57 PM	Sheet 4 of 10
File: FuelGauge.SchDoc		

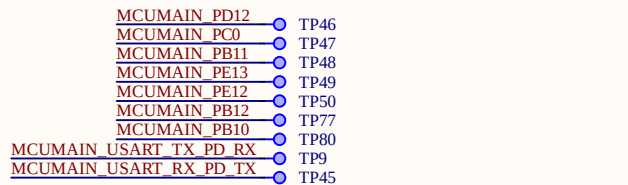


R91	R92	V (nom)	ADC (nom)
100k	0	0V	0
100k	14.3k	0.416V	681
100k	33.2k	0.832V	1363
100k	60.4k	1.248V	2044
100k	102k	1.664V	2726
100k	169k	2.08V	3407
100k	309k	2.496V	4088

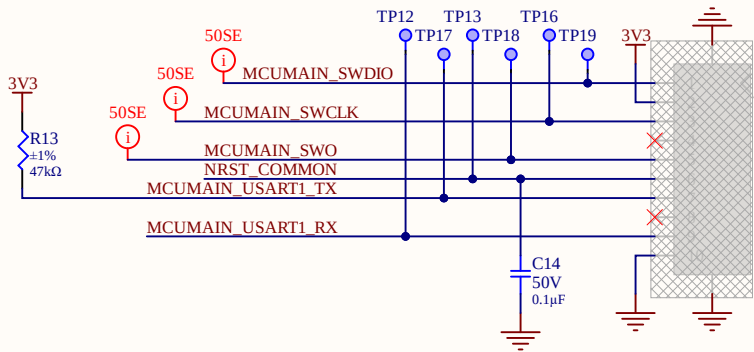
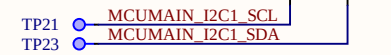


MCUMAIN_SPI1_MISO	R107	33Ω	FLASH_MISO	TP57
MCUMAIN_SPI1_MOSI	R108	33Ω	FLASH_MOSI	TP58
MCUMAIN_SPI1_SCK	R109	33Ω	FLASH_SCK	TP59
MCUMAIN_FLASH_nWP	R110	33Ω	FLASH_nWP	TP60
MCUMAIN_FLASH_nCS	R111	33Ω	FLASH_nCS	TP61
MCUMAIN_FLASH_nRESET	R112	33Ω	FLASH_nRESET	TP62

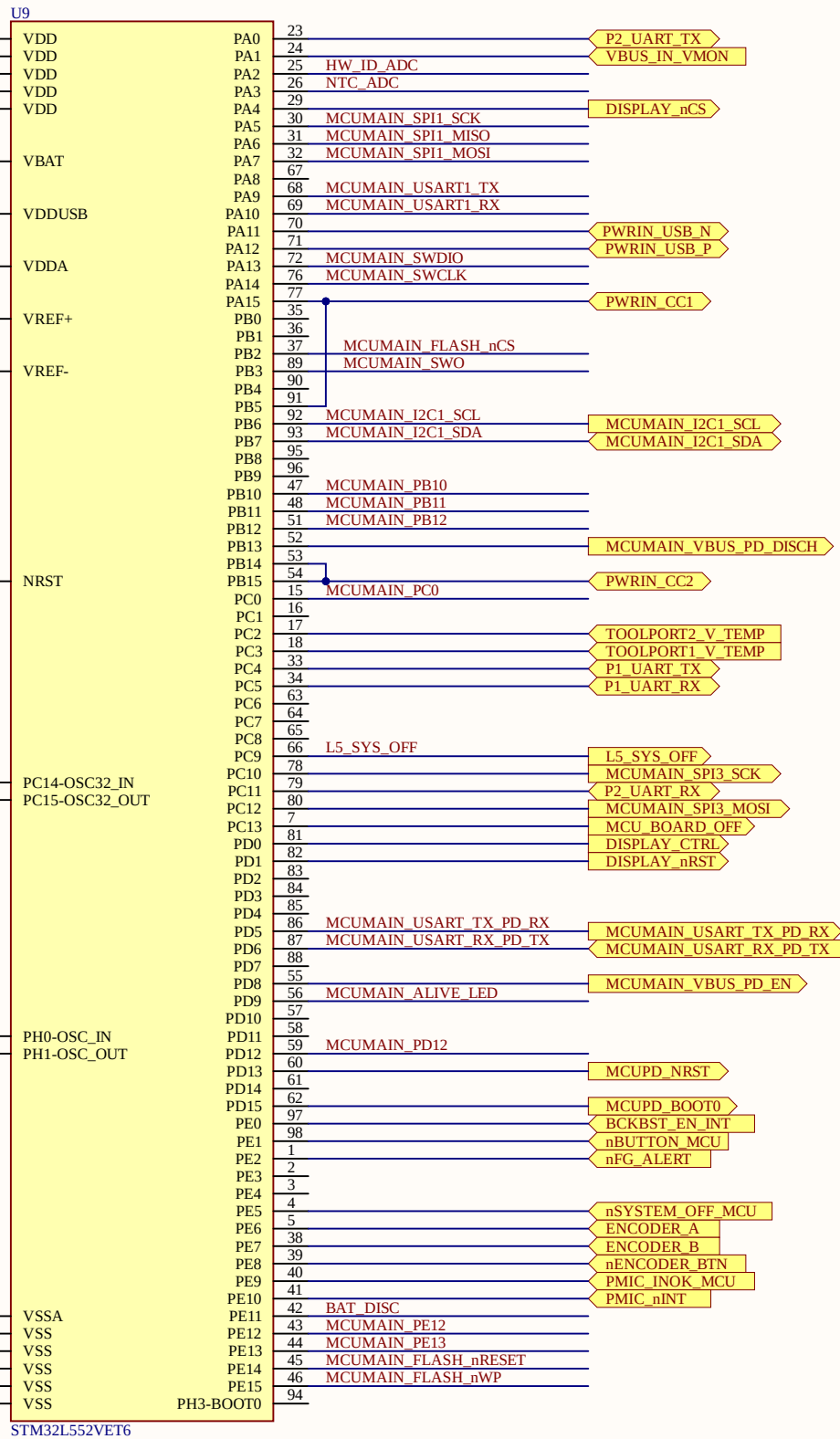
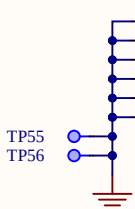
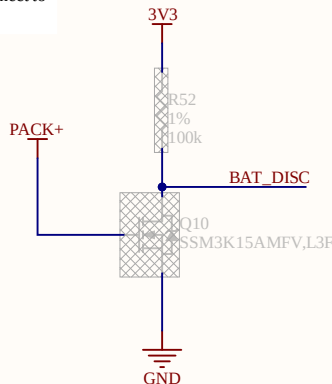
R107 shall be placed close to U17.
R108 - R112 shall be placed close to U9.



MCUMAIN_ALIVE_LED



Use Digikey part number
WM11524-ND to connect to
adapter board.



INTERRUPTS

- PE0
- PE1
- PE2
- PE3
- PE4
- PE5
- PE6
- PE7
- PE8
- PE9
- PE10
- PE11

PERIPHERALS

- PA9: USART1_TX
- PA10: USART1_RX
- PD5: USART2_TX
- PD6: USART2_RX
- PC4: USART3_TX
- PC5: USART3_RX
- PA0: UART4_TX
- PC11: UART4_RX
- PB7: I2C1_SDA
- PB6: I2C1_SCL
- PA11: D-
- PA12: D+
- PA5: SPI1_SCK
- PA7: SPI1_MOSI
- PA6: SPI1_MISO
- PB2: SPI1_NSS
- PC10: SPI3_SCK
- PC12: SPI3_MOSI
- PA4: SPI3_NSS

ADC

- ADC1_IN3
- ADC1_IN4
- ADC1_IN5
- ADC1_IN6

CONTENTS CONFIDENTIAL AND PROPERTY OF INTERCREATE AND CLIENT

Title Main MCU			 INTERCREATE 2415 3RD ST. SUITE #232 SAN FRANCISCO CA 94107
Size: A3	Number: ES-SH-2210	Revision: 10	
Date: 6/7/2024	Time: 12:49:57 PM	Sheet 5 of 10	
File: MCU_Main.SchDoc			

PC14
PC15

PA0: USART4_TX
PA1: USART4_RX

PA2: USART2_TX
PA3: USART2_RX

PA5: USART3_TX
PB0: USART3_RX

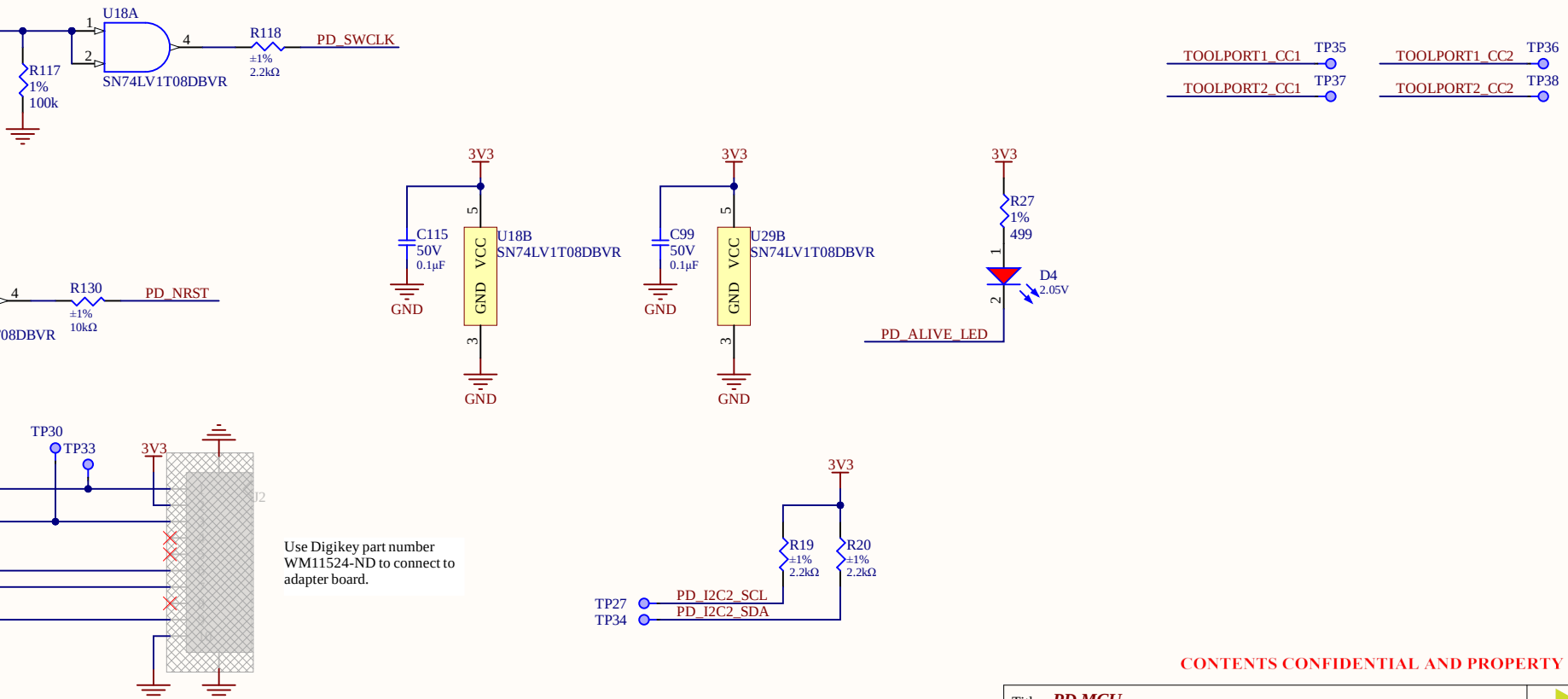
PB6: USART1_TX
PB7: USART1_RX

PA11: I2C2_SCL
PA12: I2C2_SDA

PB15: USBPD1 CC2
PA8: USBPD1 CC1
PD2: USBPD2 CC2
PD0: USBPD2 CC1

PA9: USBPD1 DBCC1
PA10: USBPD1 DBCC2
PD1: USBPD2 DBCC1
PD3: USBPD2 DBCC2

PA4: ADC1_IN4
PA6: ADC1_IN6
PA7: ADC1_IN7
PB1: ADC1_IN9

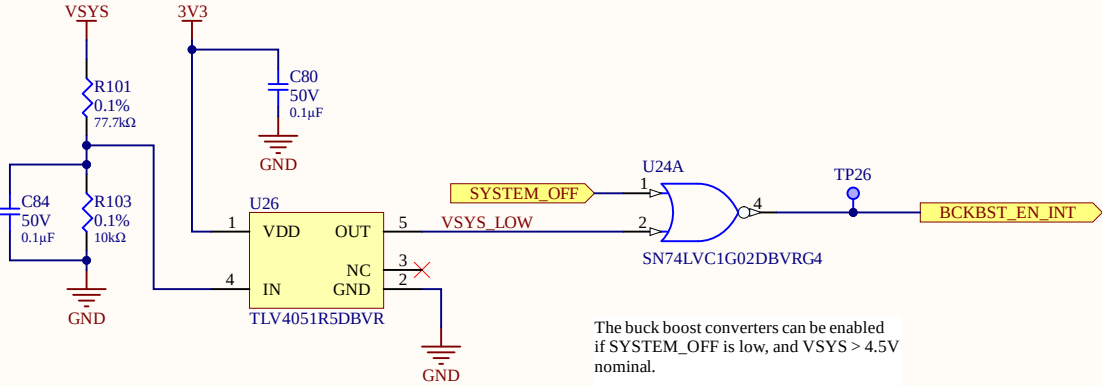


Title PD MCU			 INTERCREATE 2415 3RD ST. SUITE #232 SAN FRANCISCO CA 94107
Size: A3	Number: ES-SH-2210	Revision: 10	
Date: 6/7/2024	Time: 12:49:58 PM	Sheet 6 of 10	
File: MCU_PD.SchDoc			

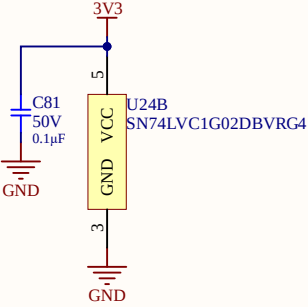


TLV4051 has a built in reference of 0.5V nominal and has an inverting push-pull output (when voltage at the IN pin is below the reference voltage, the output is high).

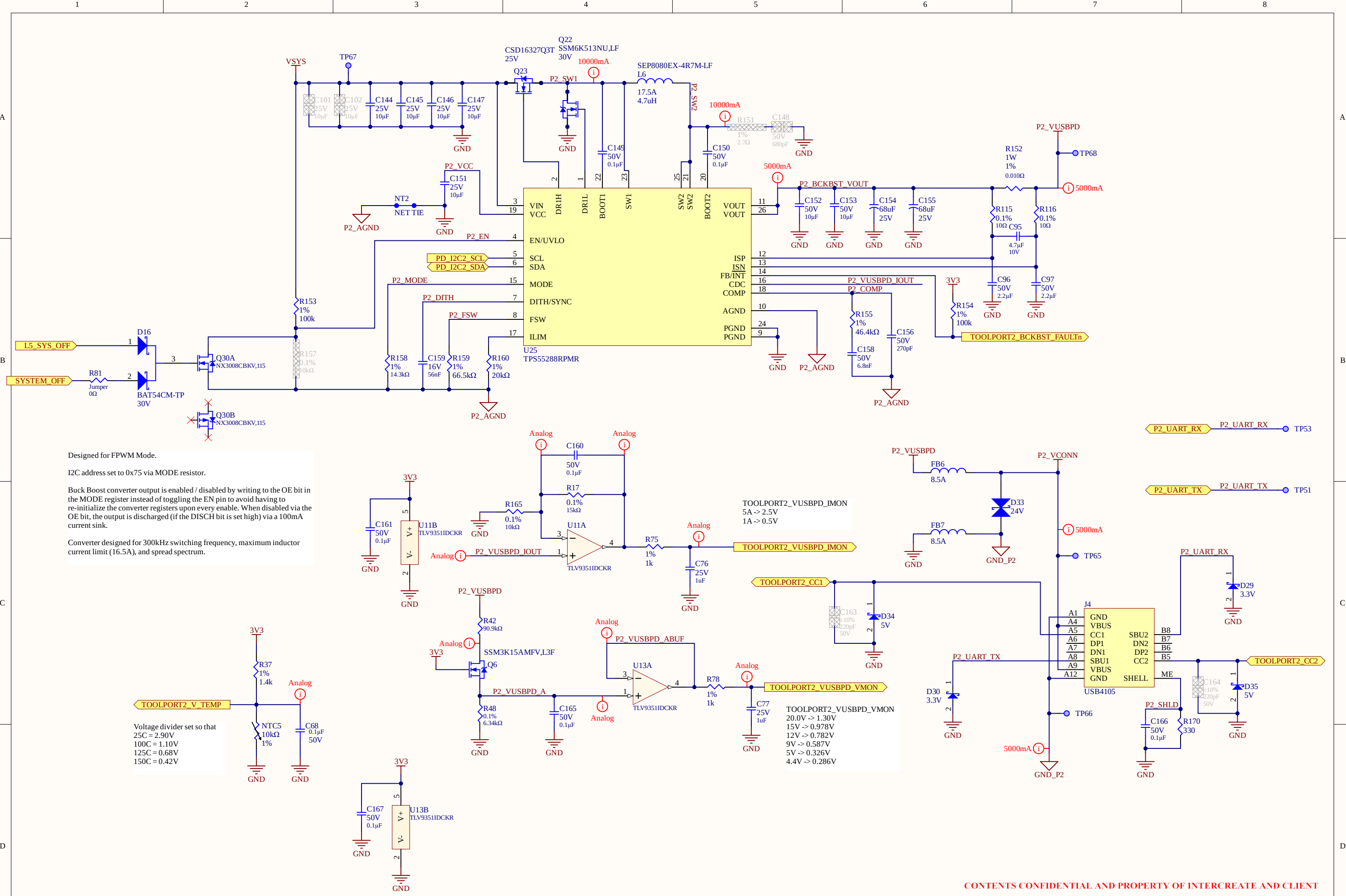
DEVICE	(V _{IT+} , V _{IT-})	OUTPUT TOPOLOGY	INPUT VOLTAGE	OUTPUT LOGIC LEVEL
TLV4051R2	0.2V, 0.18V	Push-Pull	IN > V _{IT+}	Output asserted low
TLV4051R5	0.5V, 0.48V		IN < V _{IT-}	Output asserted high
TLV4051R1	1.2V, 1.18V			

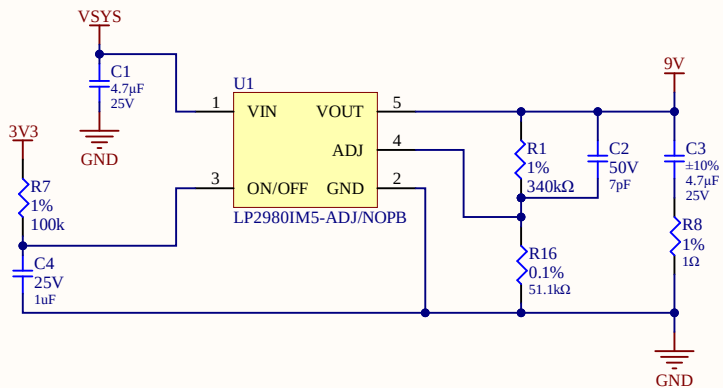
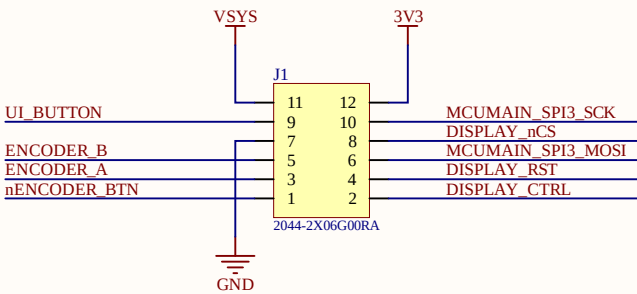


The buck boost converters can be enabled if SYSTEM_OFF is low, and VSYS > 4.5V nominal.

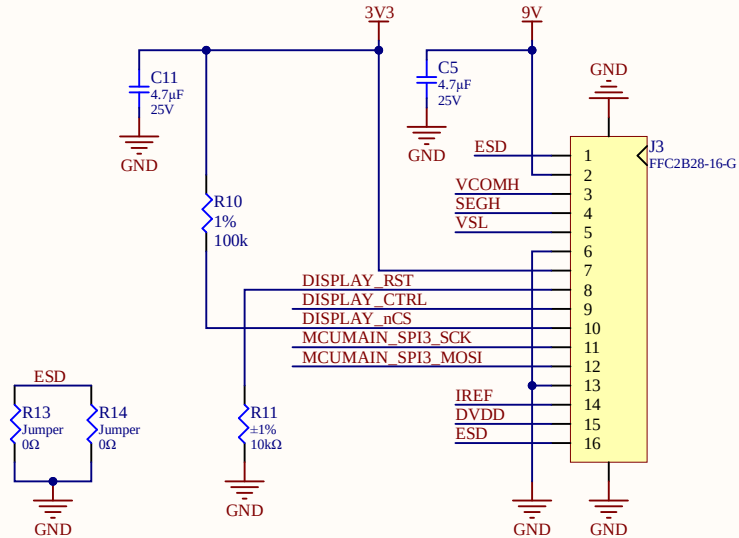
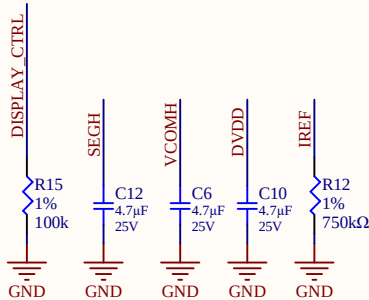
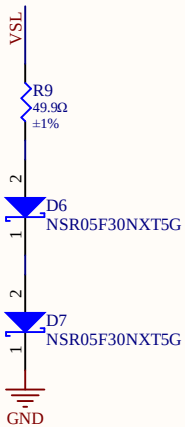
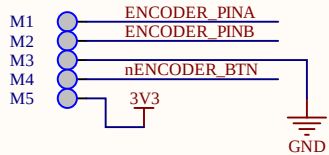
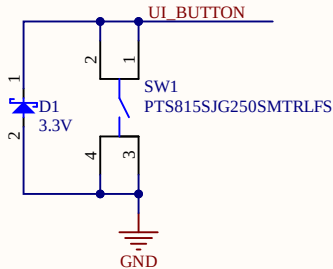


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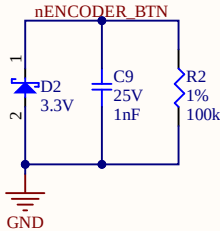
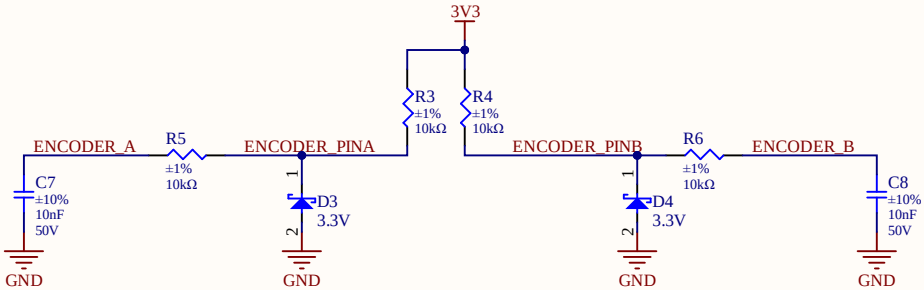




9V nominal voltage is designed for 8.84V



PIN	SYMBOL
1	ESD_GND
2	VPP
3	VCOMH
4	SEGH
5	VSL
6	VSS
7	VDD
8	RES
9	A0/SA0
10	CS
11	SCL
12	SI
13	VSS_IM1
14	IREF
15	DVDD
16	ESD_GND



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Title	UI
Size:	A3
Number:	ES-SH-2226
Revision:	4
Date:	9/1/2023
Time:	4:43:36 PM
Sheet	1 of 1
File:	UI.SchDoc

